Genomic Dissection and Prediction of Oleic Acid Concentration in Peanut Using High-Density SNP Markers

Fritzner Pierre

Crop, Soil, and Environmental Sciences Department

Final Project

Arachis hypogaea | 58K SNP Array | GWAS | Genomic Prediction | Oil Quality

Breeding Question

What genomic regions control variation in oleic acid concentration in peanut?

And how accurately can genomic prediction models predict oil quality?

1 GWAS

Identify genomic regions associated with oleic acid using FarmCPU and MLM on the 58K SNP array

2 Heritability

Estimate genomic heritability (h^2g) via GBLUP/REML using the G relationship matrix

3 Prediction

Evaluate deregressed BLUP prediction accuracy via 5-fold CV × 10 reps, reporting mean Pearson r

Background and Motivation

Why Do We Care About Oleic Acid?

Genetic Mechanism

- FAD2 enzymes convert oleic → linoleic acid
- Reduced FAD2 activity raises oleic, lowers linoleic
- Specific FAD2 allele combos enable ~80% oleic

(Lakhssassi et al. 2021; Lee et al. 2024)

Oil Quality & Stability

- Higher oleic = greater oxidative stability
- Fewer polyunsaturated fatty acids → less rancidity
- Improved frying performance & longer shelf life

(Beló et al. 2008; Lee et al. 2024)

Commercial & Breeding Value

- Healthier monounsaturated fat profile
- Suitable for high-heat industrial applications
- Conventional breeding capped at ~50–60% oleic
- Genomic tools needed to exceed this barrier

Genetic Data Description

108
Accessions
(U.S. mini core)

315
Phenotypic
observations

58K
SNP markers
(Axiom array)

1
Trait: Oleic
acid conc. (%)

Phenotypic Data

- 108 diverse accessions, U.S. peanut mini core
- 315 observations (1–4 reps, unbalanced design)
- Outlier removed: PI 429420 (studentized residual $z > 3$)
- BLUPs extracted to adjust for measurement error
- Mean raw: 53.86% oleic
- Raw SD: 7.72%

Source: PeanutBase

Genotypic Data

- Axiom_Arachis_58K SNP array
- Aligned to *A. hypogaea* cv. Tifrunner (gnm1)
- 58,233 SNPs → 10,849 after QC
- Final matrix: 104 accessions × 10,849 SNPs
- Covers cultivated *A. hypogaea* & wild diploid ancestors

Source: Legume Information System (LIS)

Analytical Pipeline

1 Data QC

- MAF ≥5%, missingness <10%
- Outlier removal (z-score)
- Genotype imputation
- BLUP extraction

2 EDA

- Histograms & descriptives
- Distribution check
- Bimodal assessment
- Phenotype normality test

3 LMM + H^2

- lmer on raw values
- Broad-sense H^2 from lmer VCs
- deBLUP/REML on BLUPs
- Genomic h^2g from G matrix

4 Pop. Structure

- PCA (PC1–PC3 covariates)
  - λ_GC ≈ 0.839 (FarmCPU)
  - λ_GC ≈ 0.941 (MLM)
  - 2 clear subgroups
- Genetic Relatedness Matrix

5 GWAS

- FarmCPU and MLM (GAPIT3)
- Manhattan + QQ plots
- Bonferroni thresholds
  - Suggestive: Chr 6, 15, & 16

6 Genomic Prediction

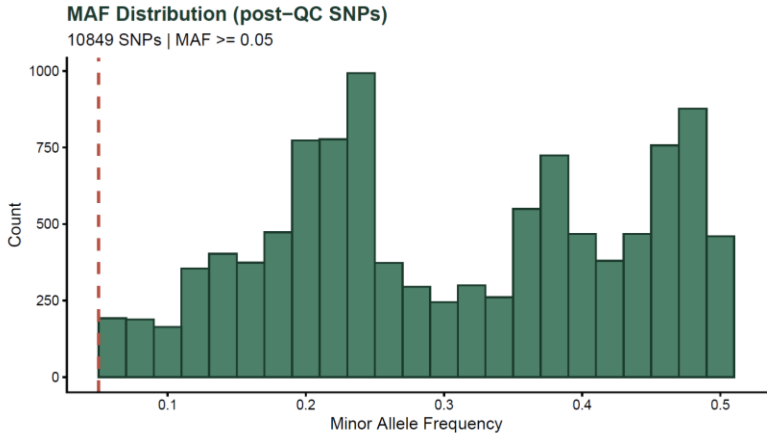
- Deregressed GBLUP
- 5-fold CV × 10 reps
  - CV mean r = 0.966
  - Overfitting gap = 0.0350

Exploratory Phenotypic Data Analysis

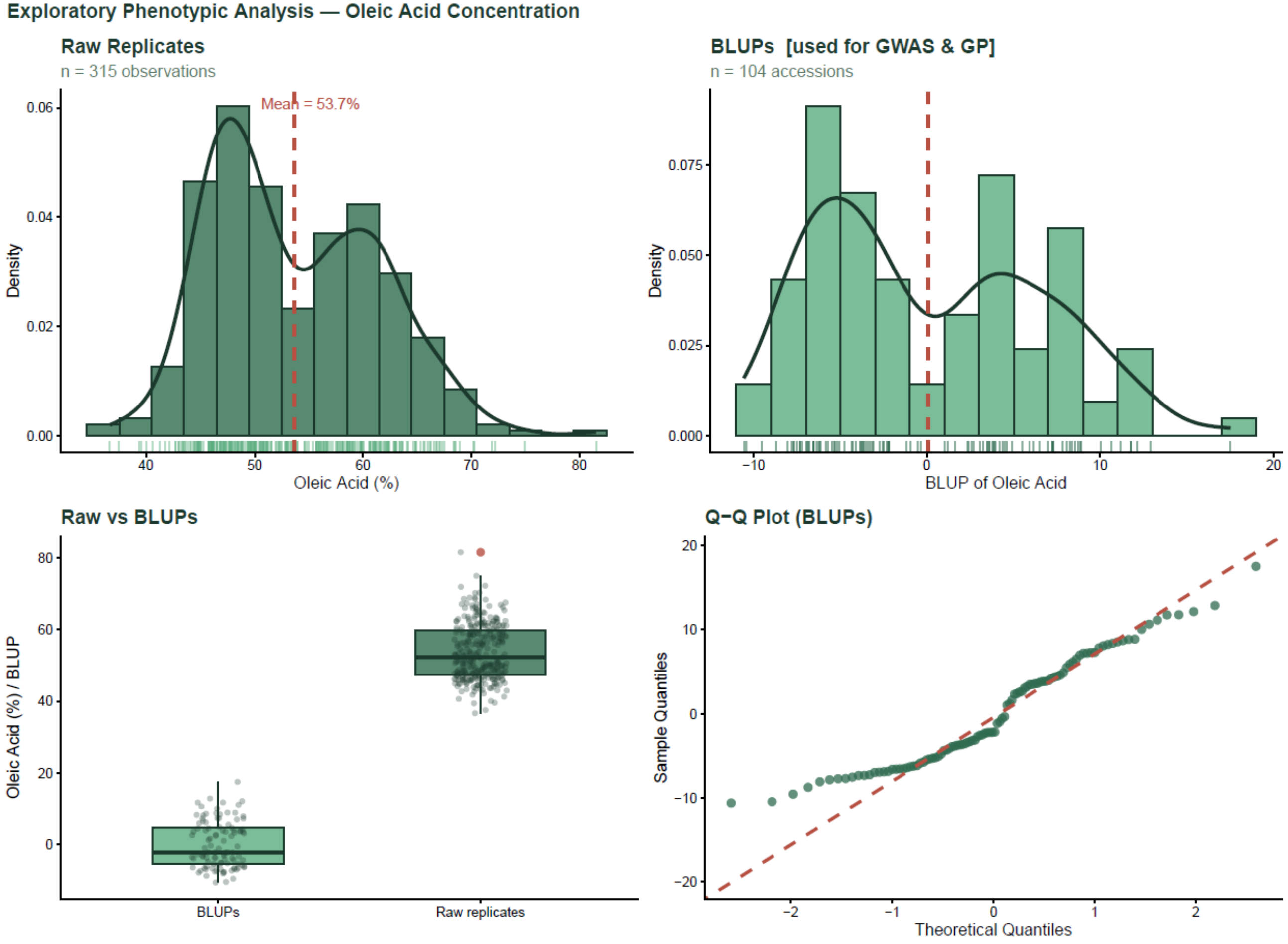
53.86%
Grand mean (108 accessions)

Bimodal
Distribution (Oleic Acid)

315 obs
1–4 reps per accession



- ✓ Removed (missingness > 10 %): 0
- ✓ Removed (MAF < 0.05): 2405
- ✓ Imputed missing: 5775
- ✓ Final matrix: 104 x 10849
- ✓ Outlier removed: PI 429420 (z > 3)
- ✓ Mean raw (%): 53.6638
- ✓ SD raw (%): 7.7207
- ✓ Mean BLUP: 0.064
- ✓ SD BLUP: 6.4594



Heritability & Population Structure

Model 1 — lmer on raw values

0.7813

Broad-sense H^2

$$H^2 = \frac{\sigma_G^2}{\sigma_G^2 + \sigma_e^2}$$

Partitions raw phenotypic variance into among-accession vs. within-accession error

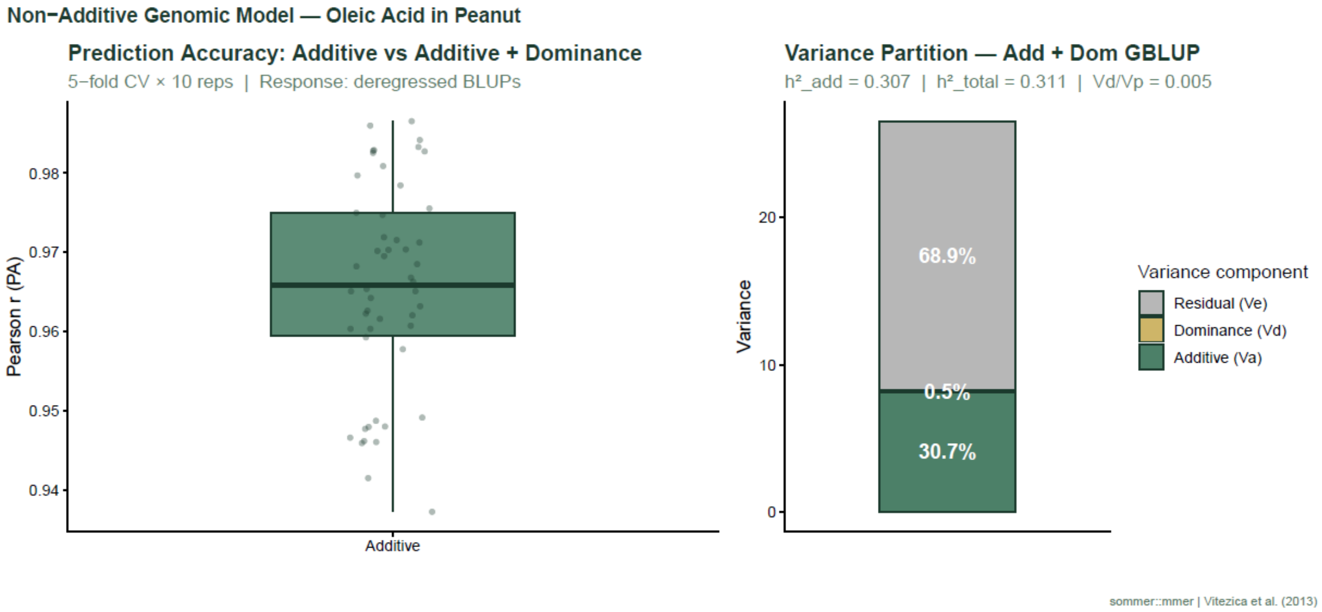
Model 2 — GBLUP on BLUPs

mixed.solve(y = BLUPs, K = G_matrix) · n = 104

σ^2g (genomic)	**8.18**
σ^2e (residual)	**18.31**
σ^2P (BLUP total)	**26.49**

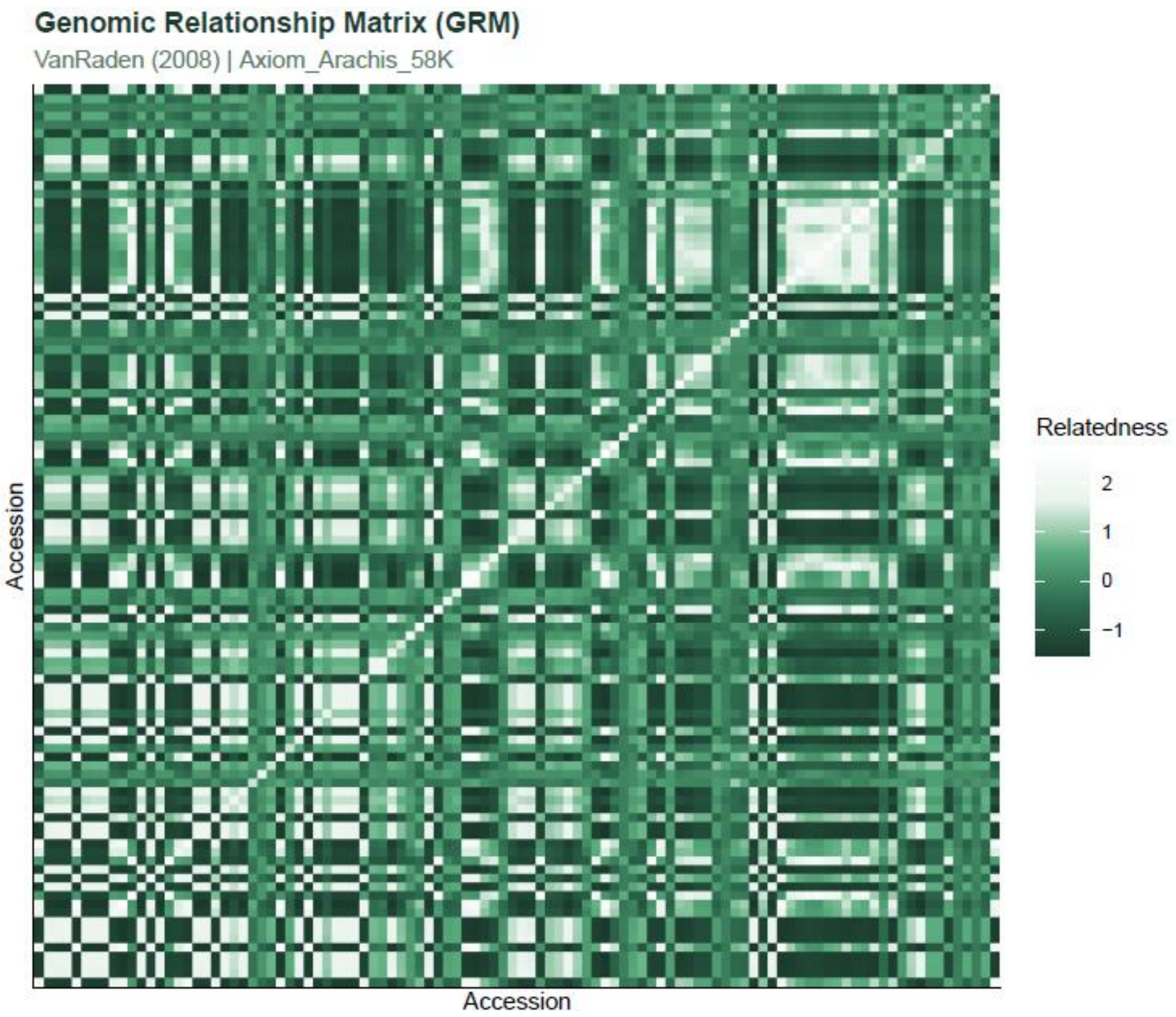
$$h_g^2 = \frac{\sigma_g^2}{\sigma_g^2+\sigma_e^2} = \frac{8.18}{8.18+18.31} = 0.3087$$ *Estimated from BLUPs*

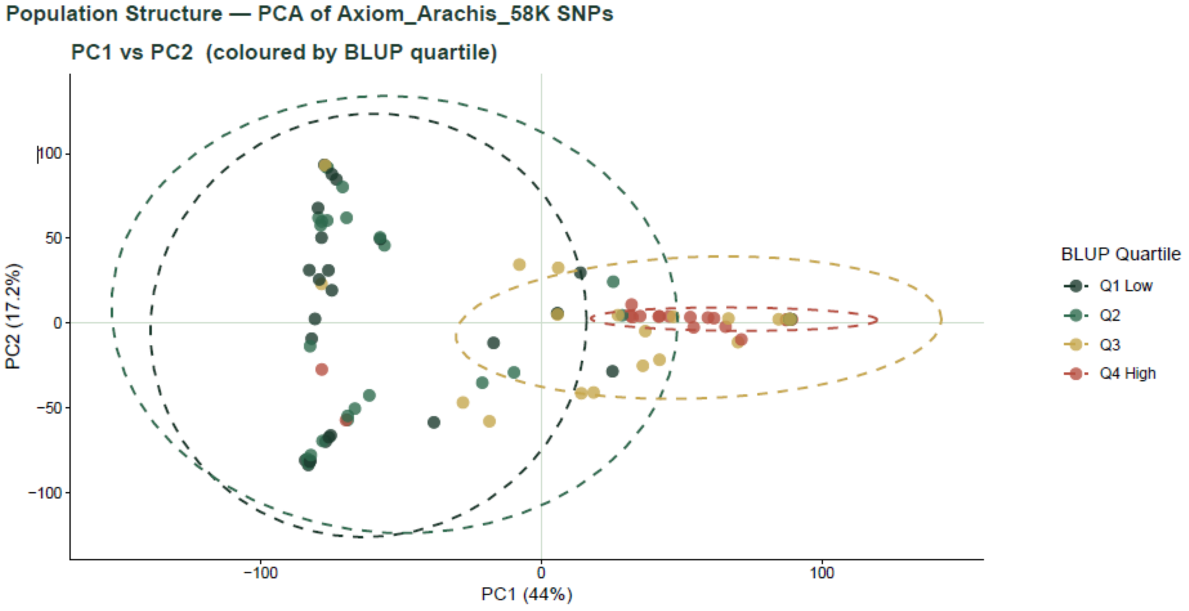
```
Gap H² - h²_g      : 0.4726

>> Gap > 0.10 - substantial non-additive variance detected.
   Fitting additive + dominance model (Step 6).
```

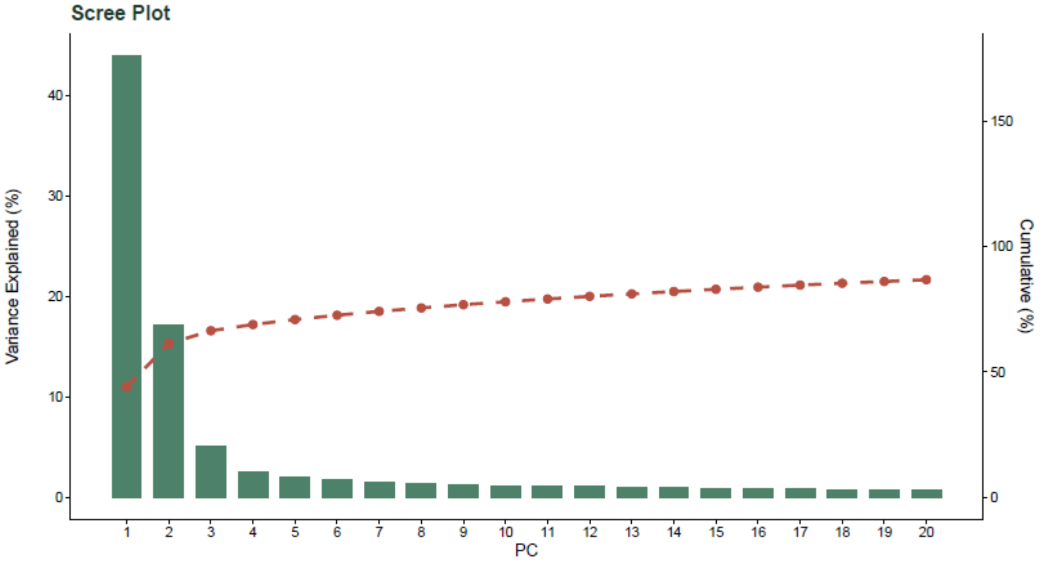



```
>> Vd/Vp = 0.0046 - dominance variance is negligible.
   Additive model is sufficient. Results retained as evidence.
```

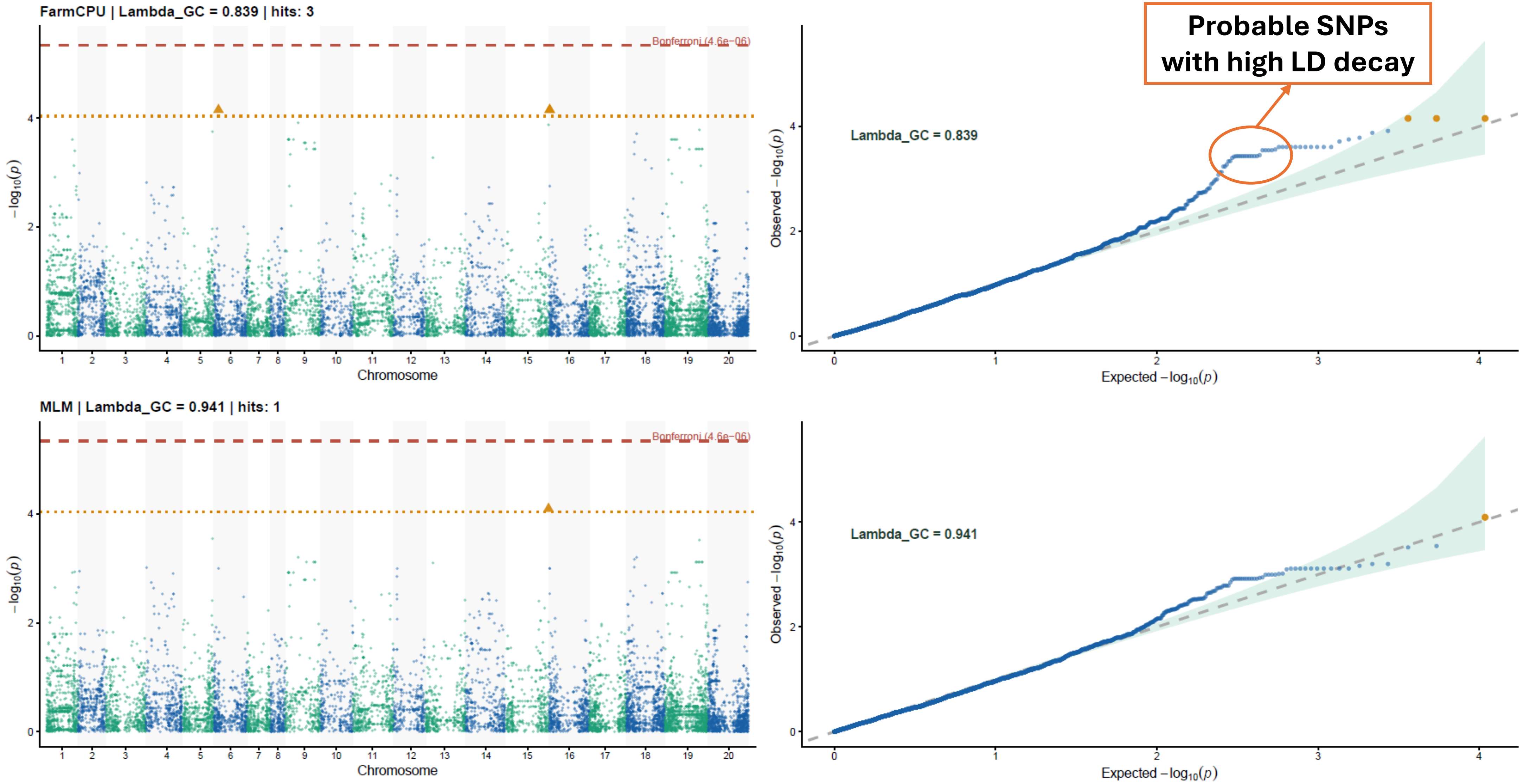
Heritability & Population Structure







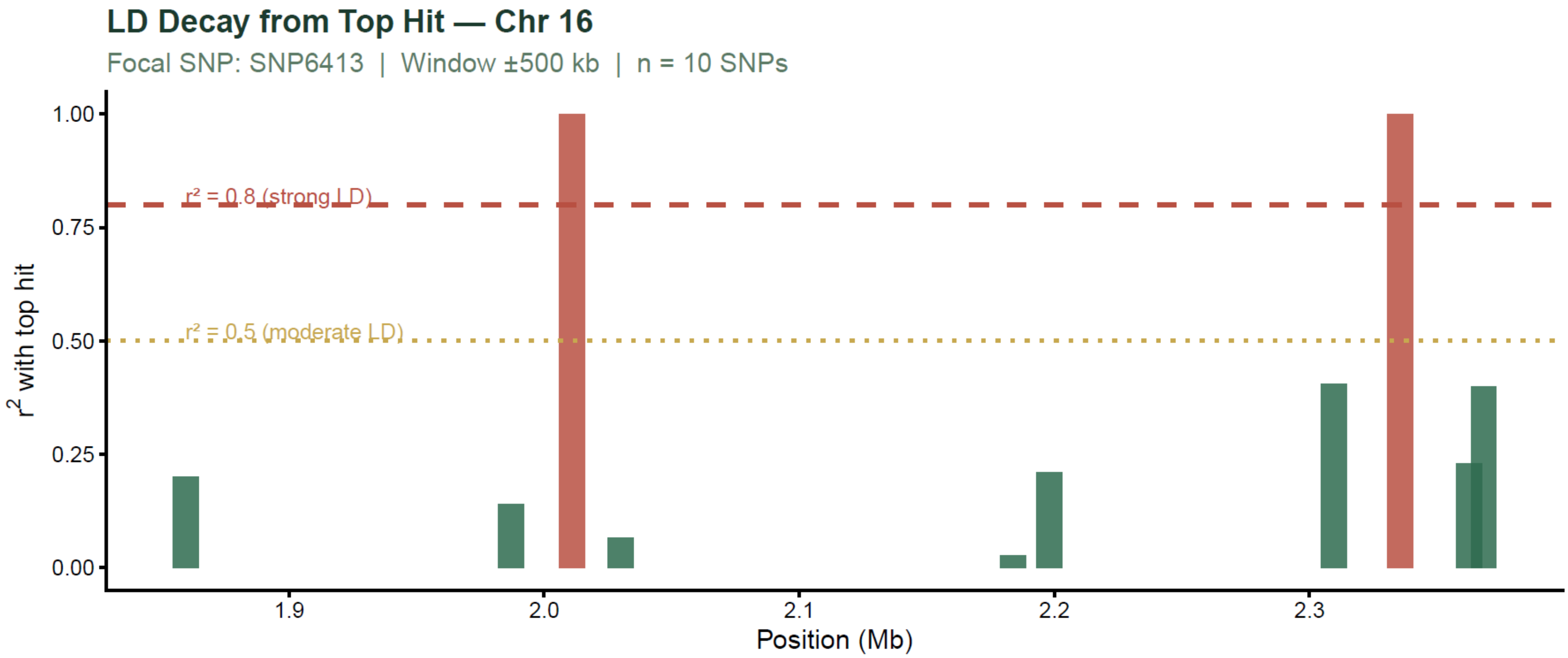
GWAS Results — Manhattan & QQ Plots



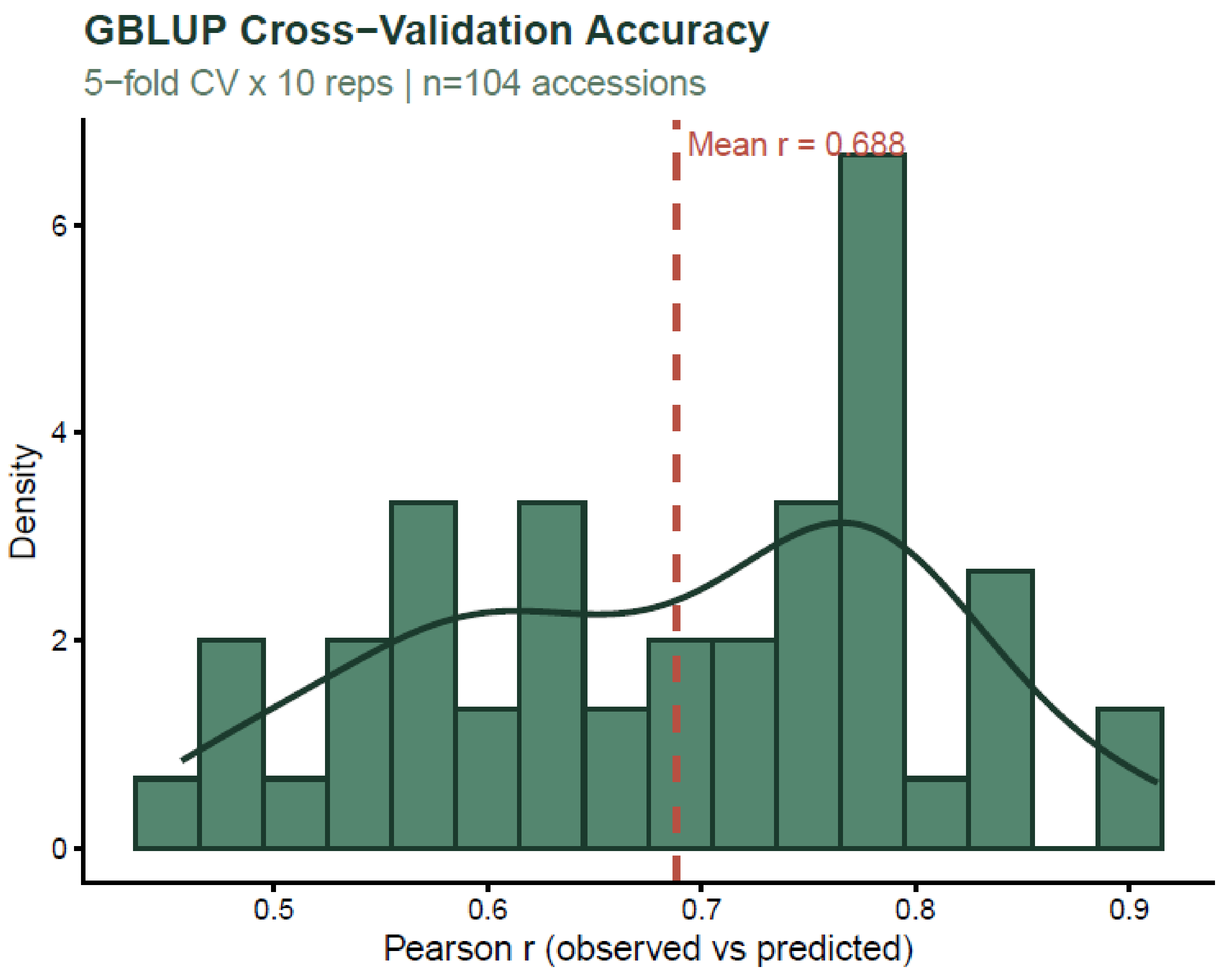
SNP	Chromosome	Position (bp)	p-value	–log10(p)	Model	Interpretation
SNP6413	16 (B06)	2,335,571	7.05E-05	4.15	FarmCPU	Suggestive
SNP2715	6 (A06)	14,862,347	7.05E-05	4.15	FarmCPU	Suggestive
SNP6408	16 (B06)	2,010,906	7.05E-05	4.15	FarmCPU	Suggestive
SNP6377	15 (B05)	147,998,830	8.07E-05	4.09	MLM	Suggestive

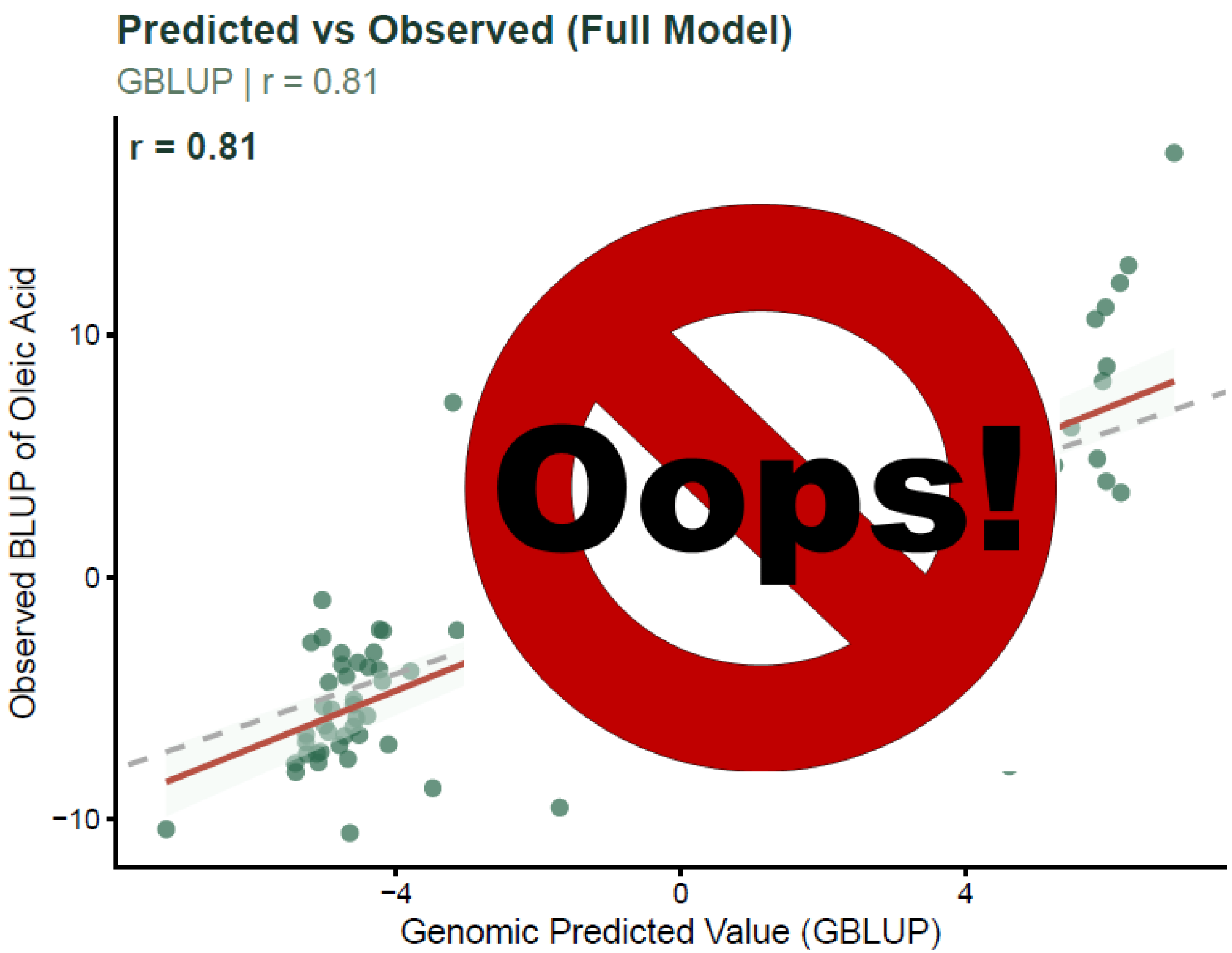
Suggestive association: Chr 6 & Chr 16 λ_GC ≈ 0.839 (genomic inflation factor) → FarmCPU pseudo-QTN approach controls for background genetic effects
$\lambda \approx 1$: population structure is well controlled (**MLM**); $\lambda > 1.1$: the test statistics may be inflated (risk of false positives); and if $\lambda < 1$: the model is conservative.

GWAS Results — Top Pair Hit Confirmation



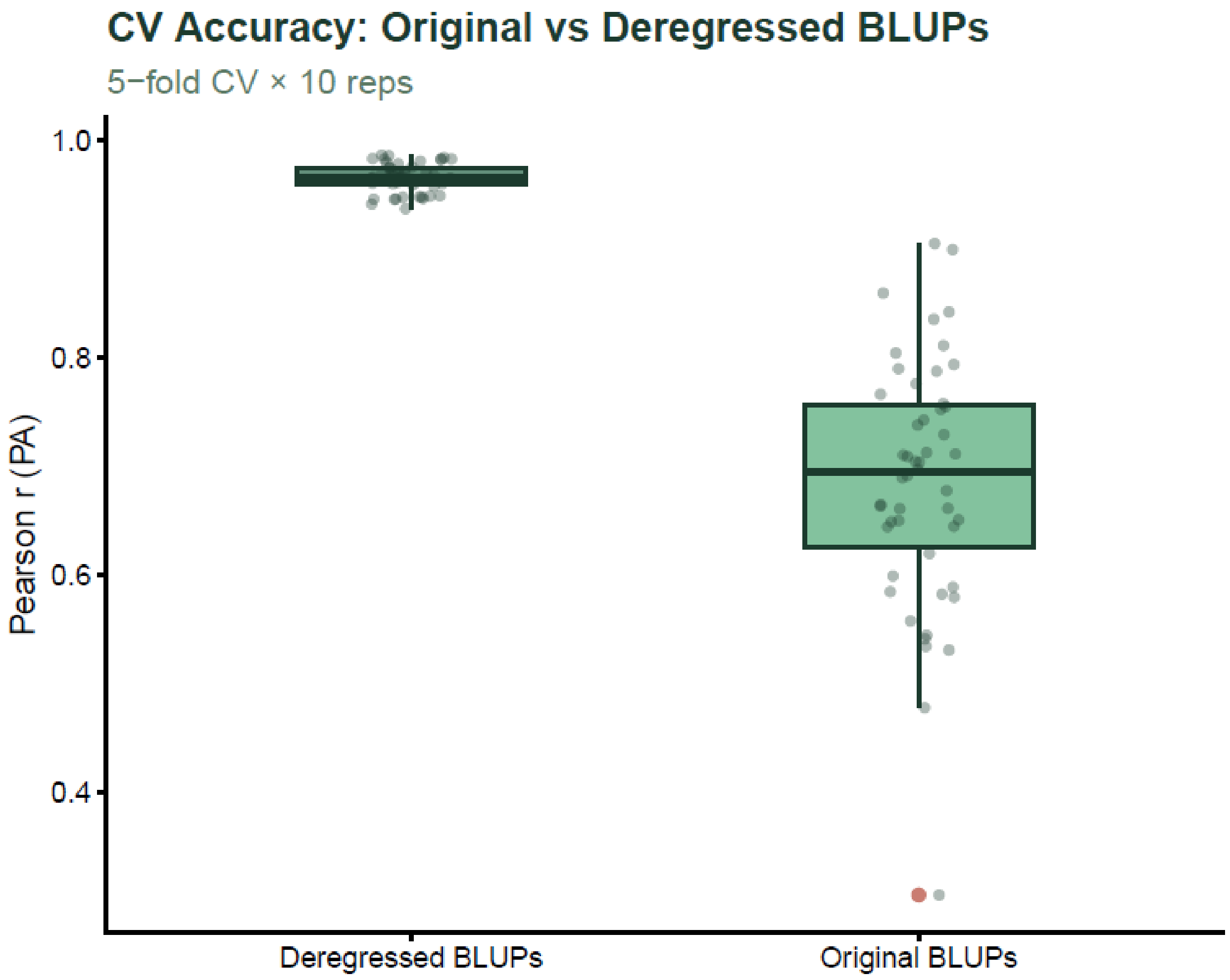
Genomic Prediction — Accuracy Metrics





DOUBLE-SHRINKAGE: BLUPs are already shrunk by lmer; passing them into a second mixed model (rrBLUP::mixed.solve) shrinks them a second time, biasing variance components downward and deflating prediction accuracy estimates.

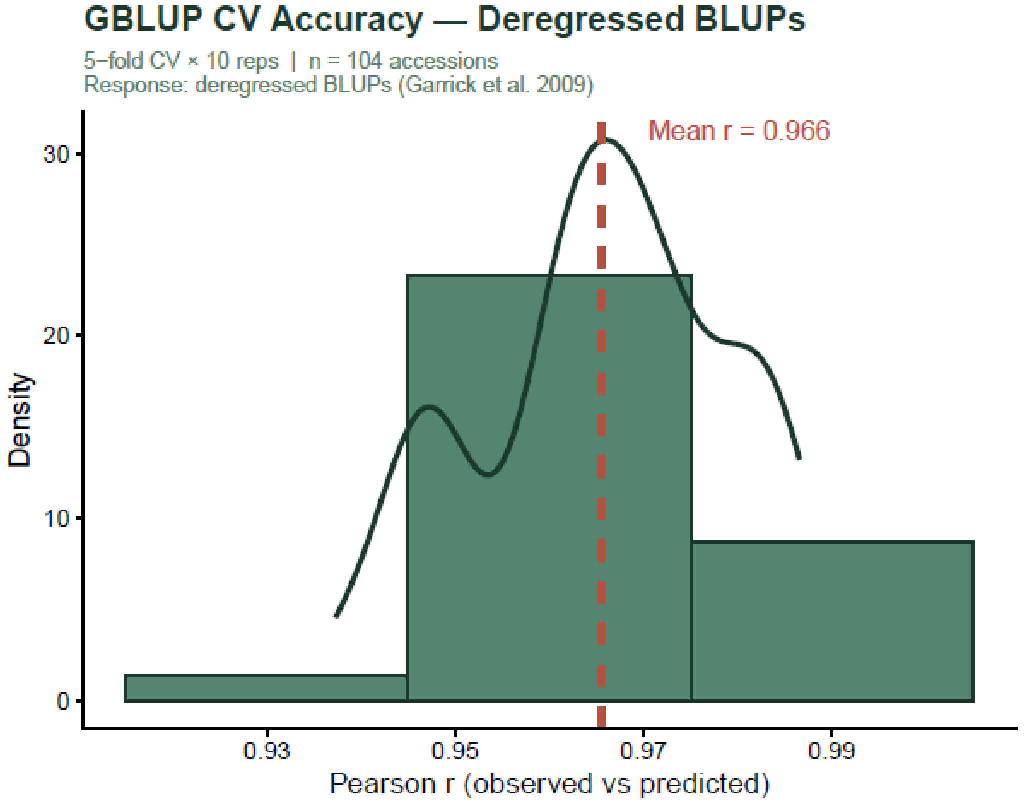
Genomic Prediction — Accuracy Metrics

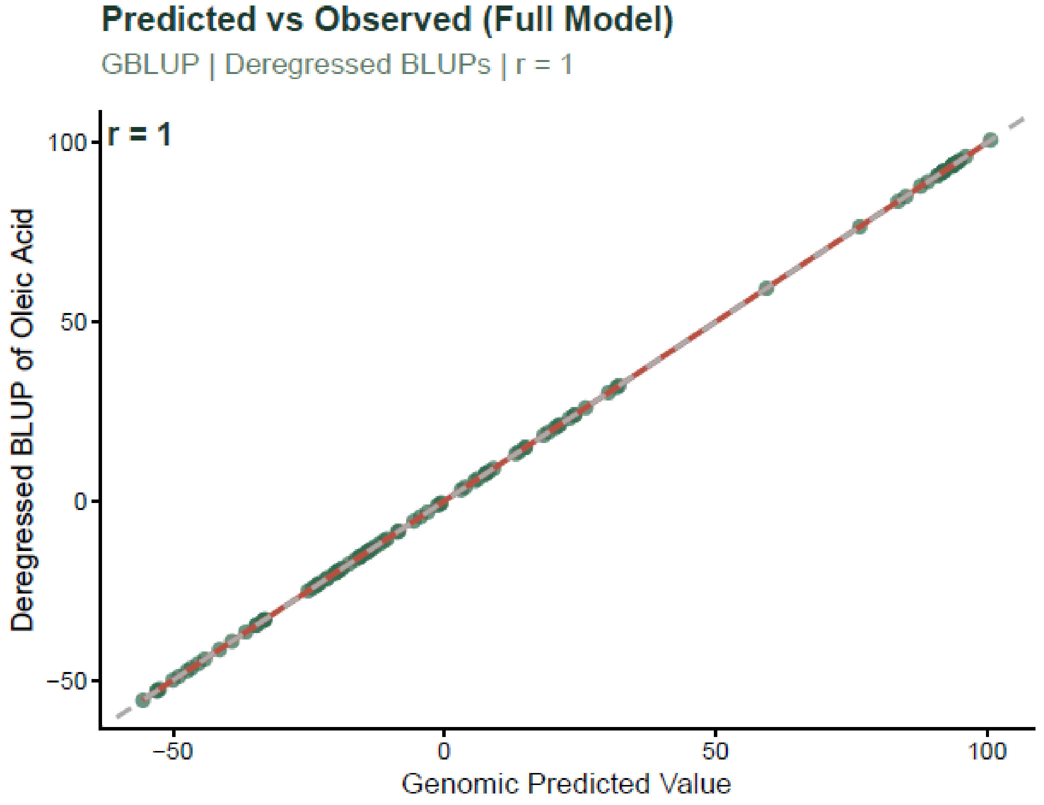


- $\tilde{g}_i = g_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}\left(0, \frac{1-r_i^2}{r_i^2}\sigma_g^2\right)$
- $g_i \rightarrow$ is the true genomic breeding value
- $\tilde{g} \rightarrow$ is the deregressed BLUP for accession i

  - $\rightarrow$is the reliability of accession i's BLUP
  - $\mathbf{D}$ *is a diagonal matrix of residual weights following Garrick et al.* (2009):
  - $d_{ii} = \frac{1-h_g^2}{\left(c+\frac{1-r_i^2}{r_i^2}\right)h_g^2}$, $\quad c = 0.1 \rightarrow$ is the proportion of genetic variance not explained by markers.

Genomic Prediction — Accuracy Metrics





Metric	Raw GBLUP	Deregressed GBLUP	Note
Mean CV r	0.686	0.9655	Pearson r across all CV folds
Spearman rho	0.8009	0.9532	Rank-based accuracy (robust to outliers)
MAE (oleic acid %)	2.8145	8.8702	Mean |predicted - observed| in % oleic
Classification accuracy	0.9135	0.9692	Threshold
SD ratio (pred/obs)	0.6924	1.0699	< 1.0 = GBLUP shrinkage toward mean
CV bias slope	1.1705	0.9017	Slope of obs ~ pred regression (ideal = 1.0)
High-oleic group r	0.2445	0.9229	Top tertile
Low-oleic group r	0.1366	0.6697	Bottom tertile

Genomic Prediction — Accuracy Metrics

5-Fold CV × 10 Reps (n=104)

0.966	[0.94,0.99]	0.9532	0.9312	8.8702%	96.92%
CV mean r (honest accuracy)	95% CI (empirical)	Spearman ρ (rank accuracy)	CV R^2 (variance explained)	MAE (oleic acid units)	Classification accuracy

Model & CV Setup

- GBLUP: $y = X\beta + Zg + e$, $g \sim N(0, G\sigma^2 g)$
  - Estimation: REML via rrBLUP mixed.solve()
- Deregressed GBLUP: $\tilde{g} = 1\mu + g + \varepsilon$, $g \sim \mathcal{N}(0, G\sigma_g^2)$, $\varepsilon \sim \mathcal{N}(0, D\sigma_e^2)$
  - Estimation: Weighted REML via rrBLUP mixed.solve()
- 5-fold CV (80% train / 20% test), 10 repetitions
- 50 independent train/test splits — stable mean r

Metric Interpretation

- CV r = 0.966 — the honest out-of-sample accuracy
- 95% CI [0.94, 0.99] — tight range confirms stability across reps
- Spearman ρ = 0.9532 — model ranks accessions correctly
- MAE = 8.87% — average prediction off by ~8.87 pp of oleic
- 96.92% correct high/low classification — key for selection decisions

Genomic Prediction — Diagnostics & Interpretation

0.9532
Spearman ρ (rank accuracy)

0.966
CV mean r (out-of-sample)

Δ 0.0350
Overfitting gap (6.88% of full r)

0.5556
Theoretical ceiling $\sqrt{h^2g}$

Overfitting

- Full r − CV r = 0.0350 (6.88%)
- Moderate — expected at n=104
- CV r is the correct metric to report

GBLUP Shrinkage

- SD ratio pred/obs = 0.692
- Predictions compressed toward mean
- CV bias slope = 0.902 (1.0 = unbiased)
- Use rankings, not raw GEBVs

Group-Level Accuracy

- High-oleic group r = 0.9229
- Low-oleic group r = 0.6697
- Prediction accuracy is higher within the high-oleic group than within the low-oleic group.
- Captures the genetic architecture of high oleic acid content better than that of low oleic acid content

The small overfitting gap (full r − CV r = 0.035, 6.88%) confirms that the deregressed GBLUP generalizes well, with minimal inflation of the full-model accuracy estimate relative to out-of-sample performance.

Integrated Results Summary

1 Trait

- Grand mean: 53.86% oleic acid
- Bimodal distribution (FAD2 segregation)
- BLUPs: mean 0.064, SD 6.46
- Strong phenotypic range supports selection

2 Heritability

- Broad-sense $H^2 = 0.7813$ (lmer, raw data)
- Genomic $h^2g = 0.3087$ (GBLUP, BLUPs)
- $h^2g < H^2$: not all causal loci fully tagged
- Two distinct models, both confirm genetic control

3 GWAS

- Suggestive peaks: Chr 6, Chr 15, & Chr 16
- Chr 16 near FAD2 locus
- $\lambda\_GC = 0.839$ (FarmCPU)
- $\lambda\_GC = 0.941$ (MLM)
- Fine-mapping needed

4 Prediction

- CV r = 0.966, 95% CI [0.943, 0.986]
- Spearman $\rho = 0.9532$ (rank accuracy)
- 96.92% high/low classification accuracy
- CV r exceeds $\sqrt{h^2g}$ ceiling (0.5556)

Conclusions & Breeding Implications

Key Conclusions

- Oleic acid is highly heritable (H^2 = 0.7813 from lmer) in this panel
- Genomic SNPs explain 31% of BLUP variance (h^2g = 0.3087, GBLUP)
- H^2 and h^2g estimated from two separate models — both valid
- Suggestive GWAS loci on Chr 6 (A06), Chr 15 (B05), and Chr 16 (B06)
- Chr 16 region consistent with known FAD2 loci in peanut
- deBLUP CV r = 0.965, Spearman ρ = 0.9532 — might be sufficient for selection
- 96.92% classification accuracy — correctly ranks high vs. low oleic
- CV r exceeds $\sqrt{h^2g}$ ceiling — FAD2 major loci boost prediction

Breeding Implications

- Genomic selection feasible — 96.92% high/low classification accuracy
- Rank accessions by GEBVs without phenotyping all lines
- Top high-oleic candidates: PI 371521, PI 370331, PI 157542
- Use rankings, not raw GEBV values (shrinkage compresses extremes)
- Chr 16 SNPs may enable marker-assisted selection for FAD2
- A larger panel would improve GWAS power & CV stability

Limitations

Small panel (n=108) limits GWAS power · No G×E partitioning

Take-Home Messages

1. A single major locus does most of the work — but not all of it. The FAD2 signal on Chr 16 explains the dominant GWAS signal and confirms that two SNPs in perfect LD tag the causal region. Yet $h_g^2 = 0.3087$ — only 40% of the broad-sense $H^2 = 0.7813$ — indicating that a substantial portion of genetic variance is not captured by the 58K array at this sample size. Genomic selection can still work well ($r \approx 0.97$) even when causal variants are not perfectly tagged, because the GRM captures genome-wide relatedness.

2. How you handle your phenotype determines what you can measure. The difference between CV $r = 0.69$ (raw BLUPs, double-shrunk) and CV $r = 0.97$ (deregressed BLUPs, correctly weighted) is not a biological difference — it is entirely a consequence of statistical methodology. Double-shrinkage is a silent bias: the model runs without errors, but the accuracy estimate is wrong. Always deregress BLUPs before using them as responses in a second mixed model.

Thank You!

```
At n=108:
> cat("  FarmCPU uses pseudo-QTN fixed effects to control polygenic background.\n")
  FarmCPU uses pseudo-QTN fixed effects to control polygenic background.
> cat("  This avoids the K-matrix confounding that hurts MLM at small n.\n")
  This avoids the K-matrix confounding that hurts MLM at small n.
> cat("  MLM with full K often over-corrects (lambda << 1) -> loss of power.\n\n")
  MLM with full K often over-corrects (lambda << 1) -> loss of power.

> cat("High-confidence loci: SNPs significant in BOTH models.\n")
High-confidence loci: SNPs significant in BOTH models.
> cat("FarmCPU-only hits: may reflect power gain from pseudo-QTN approach.\n")
FarmCPU-only hits: may reflect power gain from pseudo-QTN approach.
> cat("MLM-only hits:     rare at small n; inspect carefully for artefacts.\n\n")
MLM-only hits:     rare at small n; inspect carefully for artefacts.
```

```
H²  (broad-sense) : 0.7813  (from H2_broad_lmer.csv)
> if (!is.na(H2)) {
+   gap <- H2 - h2g_add
+   cat(sprintf("Gap H² - h²_g     : %.4f\n\n", gap))
+   if (gap > 0.10) {
+     cat(">> Gap > 0.10 - substantial non-additive variance detected.\n")
+     cat("   Fitting additive + dominance model (Step 6).\n\n")
+   } else {
+     cat(">> Gap <= 0.10 - additive model is sufficient.\n")
+     cat("   Fitting dominance model for completeness only.\n\n")
+   }
+ } else {
+   cat("H² not available - fitting non-additive model for completeness.\n\n")
+ }
Gap H² - h²_g     : 0.4726

>> Gap > 0.10 - substantial non-additive variance detected.
   Fitting additive + dominance model (Step 6).
```

```
Variance components (Additive + Dominance model):
> cat(sprintf("  sigma²_a             = %.6f\n", sigma_a2))
  sigma²_a          = 8.143355
> cat(sprintf("  sigma²_d             = %.6f\n", sigma_d2))
  sigma²_d          = 0.122529
> cat(sprintf("  sigma²_e             = %.6f\n", sigma_e2))
  sigma²_e          = 18.279312
> cat(sprintf("  Vp                   = %.6f\n", vp_ad))
  Vp                = 26.545196
> cat(sprintf("  h² (additive only)   = %.4f\n", h2_add_ad))
  h² (additive only)   = 0.3068
> cat(sprintf("  h² (add + dom)       = %.4f\n", h2_total_ad))
  h² (add + dom)       = 0.3114
> cat(sprintf("  Dominance ratio      = %.4f  (Vd/Vp)\n", dom_ratio))
  Dominance ratio      = 0.0046  (Vd/Vp)
> # Interpret dominance ratio
> if (dom_ratio > 0.05) {
+   cat(sprintf("\n>> Vd/Vp = %.4f - dominance variance is non-trivial.\n",
+               dom_ratio))
+   cat("   Non-additive model is warranted.\n\n")
+ } else {
+   cat(sprintf("\n>> Vd/Vp = %.4f - dominance variance is negligible.\n",
+               dom_ratio))
+   cat("   Additive model is sufficient. Results retained as evidence.\n\n")
+ }

>> Vd/Vp = 0.0046 - dominance variance is negligible.
   Additive model is sufficient. Results retained as evidence.
```

```
Dominance variance is negligible (Vd/Vp <= 0.05).
PA gain from adding dominance is minimal.
>> Additive model is sufficient for this dataset.
   Results above are retained as evidence.
```

```r
# -- 3.2 BLUP extraction via lmer -----------------------------------------------
cat("\nExtracting BLUPs with lmer: Value ~ 1 + (1|Taxa)...\n")
lmer_fit  <- lmer(Value ~ 1 + (1 | Taxa), data = pheno_clean, REML = TRUE)
blup_list <- ranef(lmer_fit)$Taxa
mu        <- as.numeric(fixef(lmer_fit))

blup_df <- data.frame(
  Taxa  = rownames(blup_list),
  BLUP  = blup_list[, "(Intercept)"],
  GEBV  = blup_list[, "(Intercept)"] + mu,
  stringsAsFactors = FALSE
)
cat("Grand mean (mu)       :", round(mu, 3), "%\n")
cat("BLUPs extracted for   :", nrow(blup_df), "accessions\n")
cat("BLUP range            :", round(min(blup_df$BLUP), 3), "to",
    round(max(blup_df$BLUP), 3), "\n")
write.csv(blup_df, "results/tables/BLUPs_OleicAcid.csv", row.names = FALSE)
```

```r
cat("----------------------------------------------------------------------\n")
cat("STEP 8 | Genomic Heritability\n")
cat("----------------------------------------------------------------------\n")

lmm   <- mixed.solve(y = y, K = K_mat, SE = TRUE, return.Hinv = FALSE)
Vu    <- lmm$Vu;   Ve <- lmm$Ve;   Vp <- Vu + Ve
h2g   <- Vu / Vp
cat("sigma2_g :", round(Vu, 6), "\n")
cat("sigma2_e :", round(Ve, 6), "\n")
cat("h2g      :", round(h2g, 4), "\n")
```

```r
# ==============================================================================
# STEP 10 — GENOMIC PREDICTION: GBLUP + 5-FOLD CV (10 REPS)
# ==============================================================================

cat("----------------------------------------------------------------------\n")
cat("STEP 10 | Genomic Prediction  (GBLUP, 5-fold CV x 10 reps)\n")
cat("----------------------------------------------------------------------\n")

n_total  <- length(y)
cv_store <- data.frame(rep=integer(), fold=integer(),
                       r=numeric(), RMSE=numeric(), bias=numeric())

for (rep in seq_len(N_REPS)) {
  fid <- sample(rep(seq_len(N_FOLDS), length.out=n_total))

  for (fold in seq_len(N_FOLDS)) {
    idx_t  <- which(fid == fold)
    y_cv   <- y;  y_cv[idx_t] <- NA

    fv <- tryCatch(mixed.solve(y=y_cv, K=K_mat), error=function(e) NULL)
    if (!is.null(fv)) {
      yp <- as.numeric(fv$u)[idx_t] + as.numeric(fv$beta)
      yo <- y[idx_t]
      cv_store <- rbind(cv_store, data.frame(
        rep  = rep, fold = fold,
        r    = cor(yo, yp, use="complete.obs"),
        RMSE = sqrt(mean((yo-yp)^2, na.rm=TRUE)),
        bias = coef(lm(yo ~ yp))[2]))
    }
  }
  if (rep %% 2 == 0)
    cat("  Rep", rep, "/", N_REPS, "| mean r =",
        round(mean(cv_store$r, na.rm=TRUE), 4), "\n")
}

cat("\n--- Cross-Validation Summary ---\n")
cat("Mean r   :", round(mean(cv_store$r,    na.rm=TRUE), 4), "\n")
cat("SD r     :", round(sd(cv_store$r,      na.rm=TRUE), 4), "\n")
cat("95% CI   : [",
    round(quantile(cv_store$r, 0.025, na.rm=TRUE), 4), ",",
    round(quantile(cv_store$r, 0.975, na.rm=TRUE), 4), "]\n")
cat("Mean RMSE:", round(mean(cv_store$RMSE, na.rm=TRUE), 4), "\n")
cat("Bias (slope):", round(mean(cv_store$bias, na.rm=TRUE), 4),
    "(1.0 = unbiased)\n\n")
```

```
==========================================================================
RAW BLUP MODEL
==========================================================================
Using columns: obs='observed', pred='predicted'

--- Raw BLUP GBLUP ---
Mean CV r               : 0.6860
95% CI (empirical)      : [0.4901, 0.8906]
Spearman rho            : 0.8009
MAE (oleic acid units)  : 2.8145 %
Classification accuracy : 0.9135  (threshold = 0.064%)
SD ratio (pred/obs)     : 0.6924  (1.0 = no shrinkage)
CV bias slope           : 1.1705  (1.0 = unbiased)
High-oleic group r      : 0.2445  (n=35, obs >= 3.60%)
Low-oleic  group r      : 0.1366  (n=35, obs <= -4.04%)

Saved: results/gp/prediction_diagnostics_raw.csv

==========================================================================
DEREGRESSED BLUP MODEL
==========================================================================
CV pairs loaded: 1040 observations across all folds

--- Deregressed BLUP GBLUP (CV pairs) ---
Mean CV r               : 0.9655
95% CI (empirical)      : [0.9425, 0.9856]
Spearman rho            : 0.9532
MAE (oleic acid units)  : 8.8702 %
Classification accuracy : 0.9692  (threshold = 12.881%)
SD ratio (pred/obs)     : 1.0699  (1.0 = no shrinkage)
CV bias slope           : 0.9017  (1.0 = unbiased)
High-oleic group r      : 0.9229  (n=350, obs >= 20.39%)
Low-oleic  group r      : 0.6697  (n=350, obs <= -15.63%)

Saved: results/gp/prediction_diagnostics_deregressed.csv
```

```
==========================================================================
COMBINED SUMMARY
==========================================================================

                   Metric  Value_Raw  Value_Deregressed
 95% CI lower (empirical)     0.4901             0.9425
 95% CI upper (empirical)     0.8906             0.9856
  Classification accuracy     0.9135             0.9692
            CV bias slope     1.1705             0.9017
       High-oleic group r     0.2445             0.9229
        Low-oleic group r     0.1366             0.6697
     MAE (oleic acid %)       2.8145             8.8702
                Mean CV r     0.6860             0.9655
      SD ratio (pred/obs)     0.6924             1.0699
             Spearman rho     0.8009             0.9532

Saved: results/gp/prediction_diagnostics_combined.csv

==========================================================================
Done.
```